ECE 6950

Q6.17

a) From 6.16, $\gamma = \left(1 + Q_{BO} D_{pE} / Q_{EO} \tilde{D}_{nB}\right)^{-1} = \left(1 + qAN_a x_B \tilde{D}_{pE} / Q_{EO} D_{nB}\right)^{-1}$

$A = 10^{-5}\,cm^{-2}$ $\quad \frac{Q_{EO}}{q} = 8 \cdot 10^{9}$ atom $\quad \tilde{D}_{pE} = 2\,cm^2/s$ $\quad D_{nB} = 18\,cm^2/s$ $\quad N_a = 4 \cdot 10^{17}\,cm^{-3}$ $\quad x_B = 0.5\,\mu m$

$\Rightarrow \boxed{\gamma = 0.997}$

b) $\alpha_T = 1 - \frac{x_B^2}{2 D_{nB} \tau_n}$ $\qquad \tau_n = 10^{-6}\,s$

$\Rightarrow \boxed{\alpha_T = 0.9999306}$

c) $\alpha_F = \gamma \alpha_T \Rightarrow \beta_F = \frac{\alpha_F}{1-\alpha_F} = \frac{\gamma \alpha_T}{1 - \alpha_T \gamma}$

$\alpha_F = 0.997 \Rightarrow \boxed{\beta_F = 325}$

Using $\beta_F \approx Q_{EO} \tilde{D}_{nB} / Q_{BO} = \boxed{\beta_F \approx 339} \Rightarrow \boxed{3.4\% \text{ error}}$ → pretty good approximation

Q6.18

$\beta_F = \frac{\gamma \alpha_T}{1 - \alpha_T \gamma}$, $\qquad \gamma = 0.997$ from above.

↳ Use 0.9972 for higher precision.

$\alpha_T = 1 - \frac{x_B^2}{2 D_{nB} \tau_n}$

$$\Rightarrow \beta_F = \frac{\gamma\left(1 - \frac{x_B^2}{2 D_{nB} \tau_n}\right)}{1 + \gamma\left(1 - \frac{x_B^2}{2 D_{nB} \tau_n}\right)} = \frac{\gamma\left(2 D_{nB} \tau_{no} e^{-t/t_d} - x_B^2\right)}{2 D_{nB} \tau_{no} e^{-t/t_d} + \gamma\left(2 D_{nB} \tau_{no} e^{-t/t_d} - x_B^2\right)}$$

$\tau_{no} = 10^{-6}\,s \qquad t_d = 10$

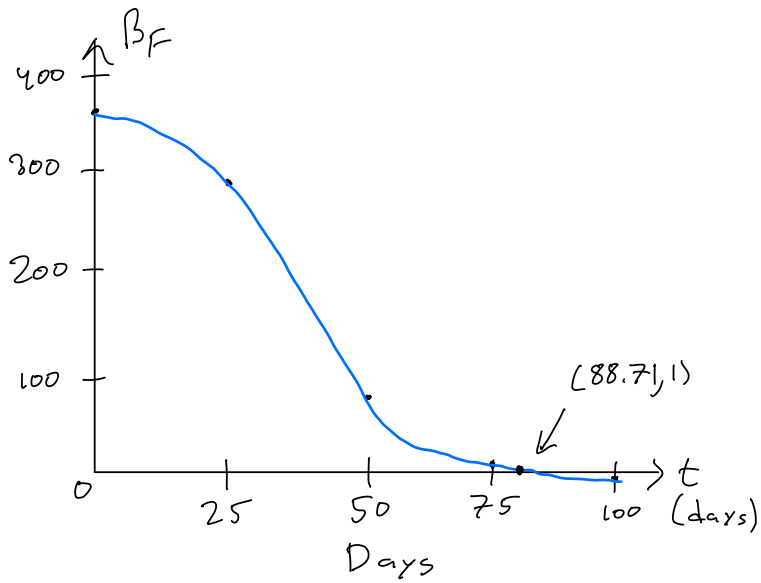
Plotting on desmos + evaluating all the many constants.

t	β_F
0	348.4
25	273.5
50	75
75	6.75
100	−0.35

$$\beta_F \propto \frac{\frac{1}{a} - e^{t/10}}{b/a + e^{t/10}}$$

$a \approx 7 \cdot 10^{-5}$

$b \approx 2.8 \cdot 10^{-3}$



By graphing calc, when $\beta_F = 1$, $t = 88.71$ days

$t =$ **88.71 days** to drop to unity

$$\tau_n = 10^{-6} \cdot e^{-\left(\frac{88.71}{10}\right)} = \tau_n = 1.4 \cdot 10^{-10} s = 0.14 \text{ nanoseconds}$$

$\tau_n = 0.14$ ns 0.14 nano

Q 6.19 We solve continuity equation,

a) Eq 5.3.10: $\frac{\partial^2 n_p}{\partial x^2} - \frac{n_p}{L_n^2} + \frac{n_{po}}{L_n^2} = 0$, $L_n^2 = D_{nB}\tau_n \Rightarrow \boxed{L_n = 13.4\,\mu m}$

From section, solution: $n_p = Ae^{-x/L_n} + Be^{x/L_n} + n_{po}$ BC: $0 = Ae^{-x_B/L_n} + Be^{x_B/L_n} + n_{po}$

$\Rightarrow n_p(0) = A(1-e^{-x_B/L_n}) + B(1-e^{x_B/L_n}) = A + B + n_{po}$

By Wolfram solve for coeffs,

$$A = \frac{(n_p(0)-n_{po})e^{x_B/L_n} + n_{po}}{\underbrace{e^{x_B/L_n} - e^{-x_B/L_n}}_{2\sinh(\frac{x_B}{L_n})}} \qquad B = \frac{(n_{po}-n_p(0))e^{-x_B/L_n} - n_{po}}{\underbrace{e^{x_B/L_n} - e^{-x_B/L_n}}_{2\sinh(\frac{x_B}{L_n})}}$$

flip/negate

$$\Rightarrow \boxed{n_p = n_{po} + \frac{(n_p(0)-n_{po})e^{x_B/L_n} + n_{po}}{2\sinh(\frac{x_B}{L_n})}e^{-x/L_n} + \frac{(n_{po}-n_p(0))e^{-x_B/L_n} - n_{po}}{2\sinh(\frac{x_B}{L_n})}e^{x/L_n}}$$

b)

$$I_{recomb} = (AJ_{nC} - AJ_{nE}) \quad J_{nC} = qD_{nB}\frac{\partial n_p}{\partial x}\Big|_{x_B} \quad J_{nE} = qD_{nB}\frac{\partial n_p}{\partial x}\Big|_0$$

(I_r)

$$\Rightarrow \frac{\partial n_p}{\partial x} = -\frac{(n_p(0)-n_{po})e^{x_B/L_n} + n_{po}}{2\sinh(\frac{x_B}{L_n})L_n}e^{-x/L_n} + \frac{(n_{po}-n_p(0))e^{-x_B/L_n} - n_{po}}{2\sinh(\frac{x_B}{L_n})L_n}e^{x/L_n}$$

By calculator and noting $n_p(0) = n_{po}e^{qV_{BE}/kT}$

$$AJ_{nE} = qD_{nB}A\frac{\partial n_p}{\partial x}\Big|_0 = qD_{nB}A\frac{(n_{po}-n_p(0))\cosh(\frac{x_B}{L_n}) - n_{po}}{L_n\sinh(\frac{x_B}{L_n})}$$

(A is area)

$$AJ_{nC} = qD_{nB}A\frac{\partial n_p}{\partial x}\Big|_{x_B} = qD_{nB}A\frac{(n_{po}-n_p(0)) - n_{po}\cosh(\frac{x_B}{L_n})}{L_n\sinh(\frac{x_B}{L_n})}$$

Collecting,

$$I_{rB} = AJ_{nC} - AJ_{nE} = \frac{qD_{nB}A}{L_n\sinh(\frac{x_B}{L_n})}\left[\begin{array}{l}(n_{po}-n_p(0)) - n_{po}\cosh(\frac{x_B}{L_n}) - \\ (n_{po}-n_p(0))\cosh(\frac{x_B}{L_n}) + n_{po}\end{array}\right]$$

$$\boxed{I_{B\text{-recomb}} = \frac{qD_{nB}A}{L_n\sinh(\frac{x_B}{L_n})}\Big(2n_{po} - \underbrace{n_{po}e^{qV_{BE}/kT}}_{n_p(0)}\Big)\Big(1 - \cosh(\frac{x_B}{L_n})\Big)}$$

c) $\gamma = \dfrac{1}{1 + I_{pE}/I_{nE}} = \dfrac{1}{1 + J_{pE}/J_{nE}}$

Define constants from table + online + question

$N_{aB} = 10^{17}\ cm^{-3}$ $\quad D_{nB} = 20\ \frac{cm^2}{s}$ $\quad \tau_n = 10^{-7} s$ $\quad x_E = x_B = 1\mu m$

$N_{dE} = 10^{19}\ cm^{-3}$ $\quad D_{pE} = 2\ \frac{cm^2}{s}$ $\quad \tau_p = 10^{-9} s$ $\quad L_n = (D_n \tau_n)^{1/2}$

$L_p = (D_p \tau_p)^{1/2}$

(At forward bias)

From above, $J_{nE} \propto D_{nB} \dfrac{(\cancel{n_{po}} - n_p(0)) \cosh\left(\frac{x_B}{L_n}\right) - \cancel{n_{po}}}{L_n \sinh\left(\frac{x_B}{L_n}\right)} \approx \dfrac{D_{nB}\, n_i^2 e^{\frac{qV_{BE}}{kT}} \cosh\left(\frac{x_B}{L_n}\right)}{N_{aB}\, L_n \sinh\left(\frac{x_B}{L_n}\right)}$

$\Rightarrow J_{pE} \propto \dfrac{D_{pE}\, n_i^2 e^{\frac{qV_{BE}}{kT}} \cosh\left(\frac{x_E}{L_p}\right)}{N_{dE}\, L_p \sinh\left(\frac{x_E}{L_p}\right)}$ (just swap constants)

$$\Rightarrow \gamma = \frac{1}{1 + J_{pE}/J_{nE}} = \left(1 + \frac{\left(\dfrac{D_{pE} \cosh\left(\frac{x_E}{L_p}\right)}{N_{dE}\, L_p \sinh\left(\frac{x_E}{L_p}\right)}\right)}{\left(\dfrac{D_{nB} \cosh\left(\frac{x_B}{L_n}\right)}{N_{aB}\, L_n \sinh\left(\frac{x_B}{L_n}\right)}\right)}\right)^{-1}$$

Using constants defined above, by calculator,

$$\boxed{\gamma = 0.9977}$$

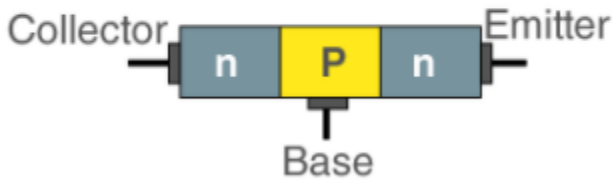
$\alpha_F = \gamma \alpha_T \Rightarrow \alpha_T = \dfrac{J_{nC}}{J_{nE}} = \left(\text{Both defined above in b)}\right) = 0.9972$ by calc

$\alpha_F = \gamma \alpha_T = \alpha_F = 0.9949$

$\beta_F = \dfrac{\alpha_F}{1 - \alpha_F} = \boxed{\beta_F = 195.08}$

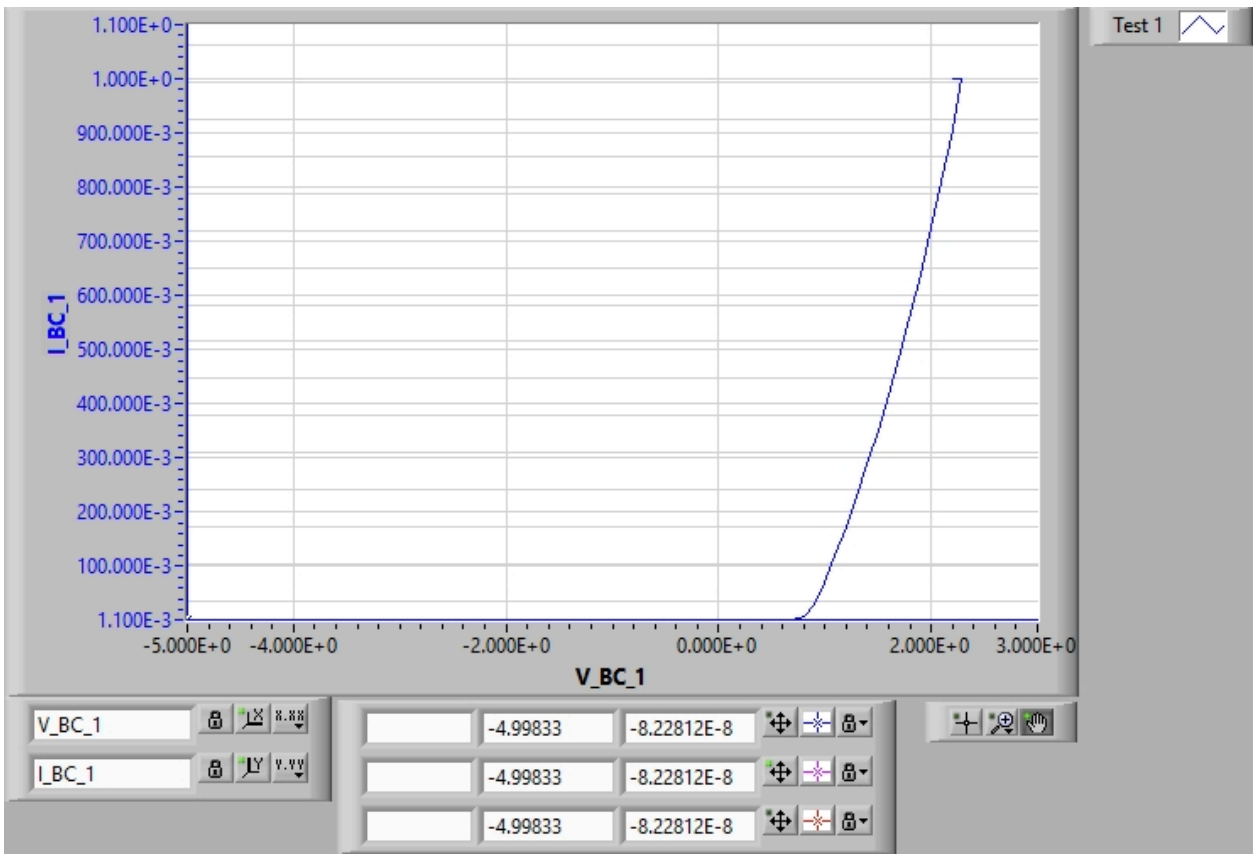
Bryant Har
Lab 6

For a diagram



Q1. Measure and plot the two-terminal I −V curves of the base-emitter, base-collector, and collector-emitter junctions with the third terminal floating. Explain the behavior.

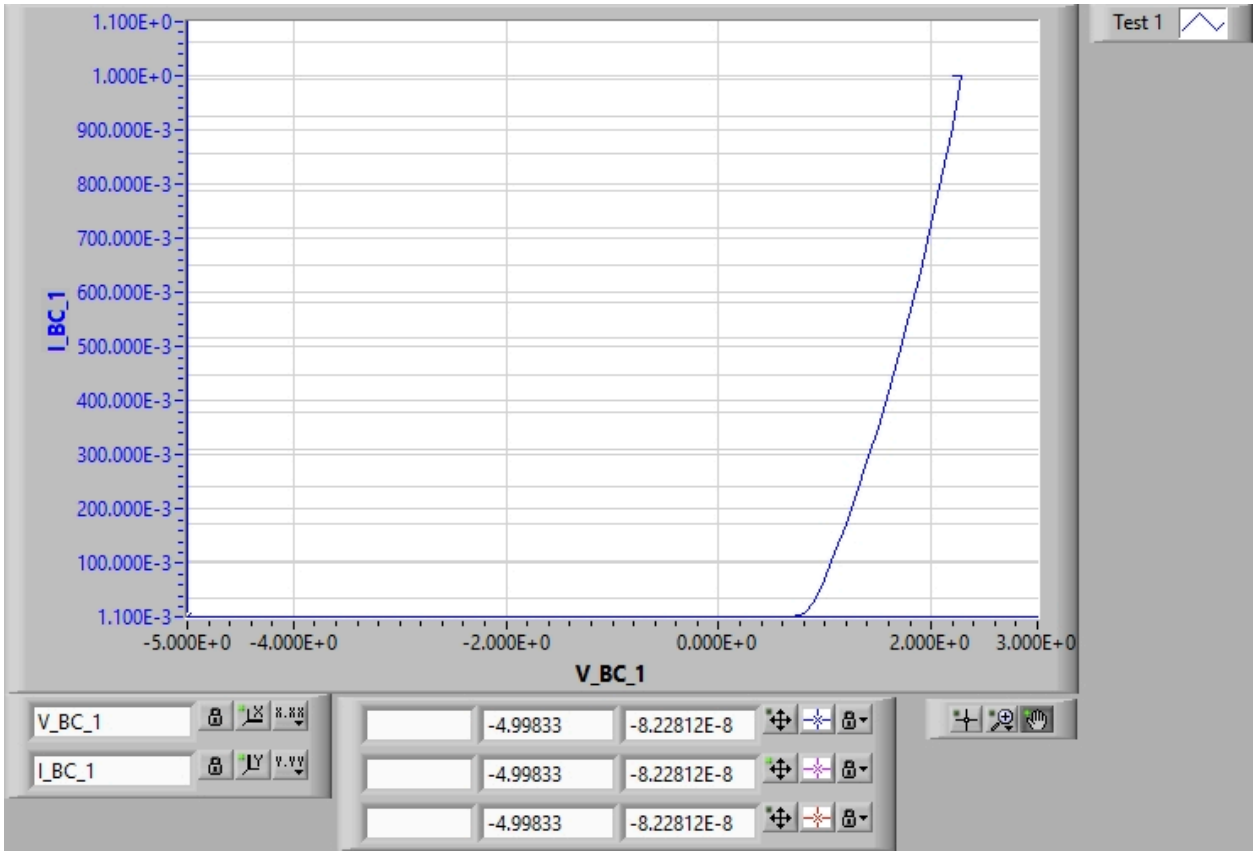
Base-Emitter Plot (Mislabeled Axes; Should be “BE”, not “BC”)



Explanation:

As we see in the diagram, there will be a potential developed between the base and the emitter. The diode starts to turn on at 0.7V. For positive voltage difference, this will cause electrons to flow from the emitter to the base and holes vice versa. This will cause current flow to increase per the diode equation.

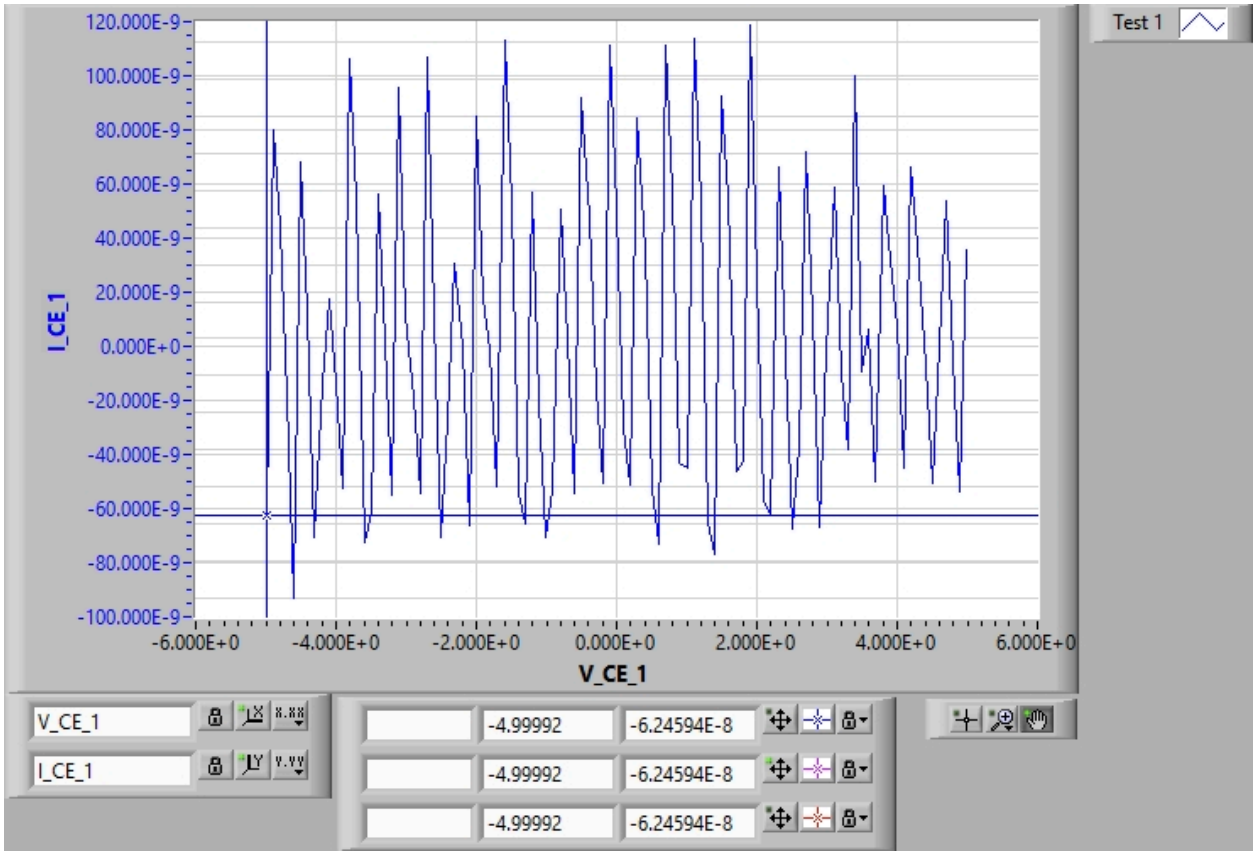
Base-Collector Plot



Explanation:

As we see in the diagram, this is basically the mirrored situation of the top. As such, the curve is very similar. There will be a potential developed between the base and the collector. The diode starts to turn on at 0.7V. For positive voltage difference, this will cause electrons to flow from the emitter to the base and holes vice versa. This will cause current flow to increase per the diode equation. The major difference between this case and the above case is the doping of the n-type semiconductor.

Collector-Emitter Plot

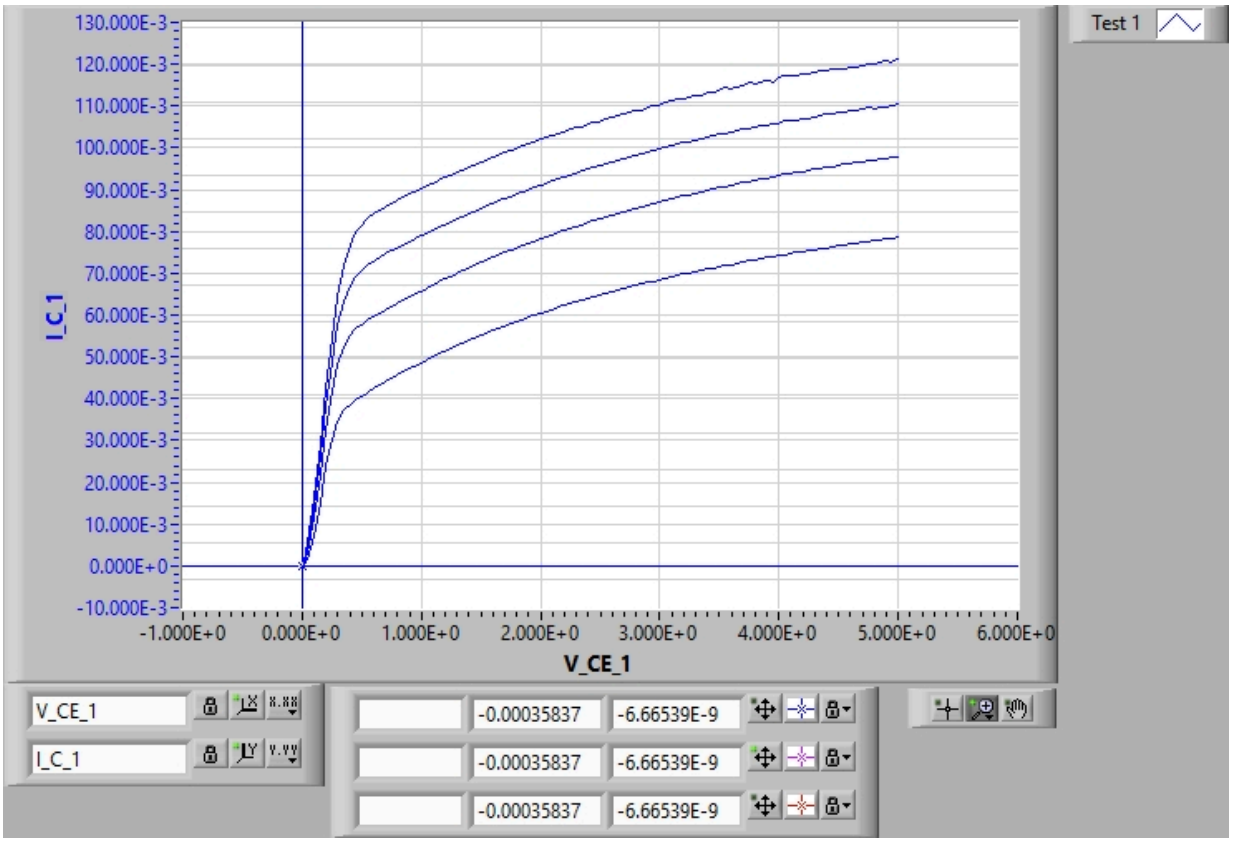


Explanation:

Here, we create a voltage difference between the collector and the emitter. As we would expect, no current flows (just noise around I=0). This is because there is a p layer between the two, and as such, current cannot flow as there is no applied voltage to the base layer. With no applied voltage to the base layer, nothing (i.e. electrons) can get through the p-layer. Intuitively, it's like two pn-diodes sandwiched with a voltage difference on the same end (the n-type terminal). Clearly, nothing can flow between the two terminals due to the shared p-layer.

Q2. Measure and plot the IC vs. VCE as you step through various base currents IB to generate the output characteristics of the Silicon bipolar junction transistor (BJT). Explain the behavior, and estimate the Early voltage.

Ic vs Vce Plot



Explanation:

We notice that the current increases exponentially before saturating and tapering off. As you increase the current going through the base, there are more holes supplied through the base, and therefore more current can flow. That is why each curve is higher than the last. There is more current supplied to the base, so the resultant current is higher. The shape of each curve itself is the classic transistor curve. The early voltage should correspond to the x-axis intercept of the linearized part of the curve after it has tapered off. The cutoff region is the area below the lowest curve, and the saturation region relates to the linear ramp-up section and the area above that. The area where each curve flattens out is the forward active region.

From graph, we mark two points for each curve from top to bottom. From this, we linearize the linear portion of each curve.

Early Voltages (For each curve from top to bottom)

	Linearized Curve	Early Voltage (x-intercept)

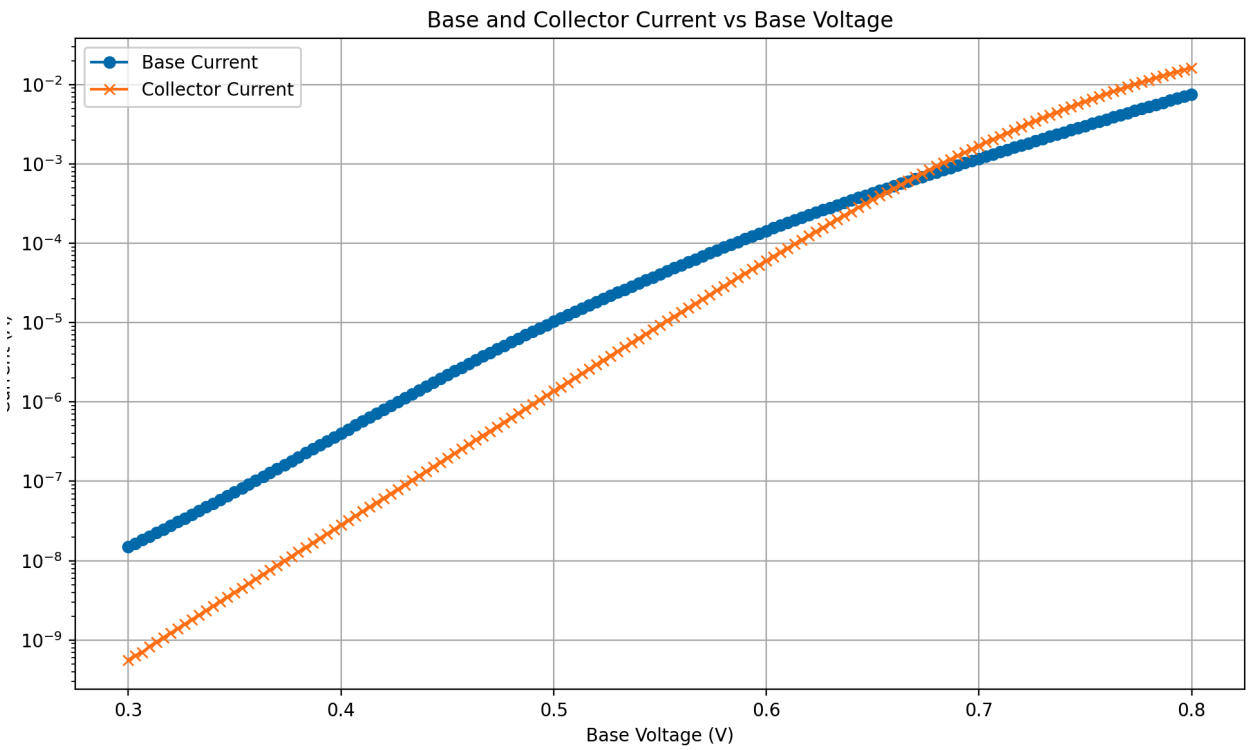
Top Curve	y= 0.005x + 0.095	-19V
Second from Top Curve	y= 0.0052x + 0.08	-15.38V
Second from Bottom Curve	y= 0.004x + 0.065	-16.25V
Bottom Curve	y= 0.0046x + 0.05	-10.87V

Averaging these, we get

EARLY VOLTAGE = -15.375V

Q3. Consider Figs 7.3 and 7.4 of MKC. For the device you measured, show the Gummel plot (Transfer curve) of the BJT which has the collector current (IC) and base current (IB) in log-scale vs the base-emitter voltage VBE , and the current gain β = IC /IB vs log(IC) similar to the 14 Homework 6 figures in MKC. Explain the behavior.

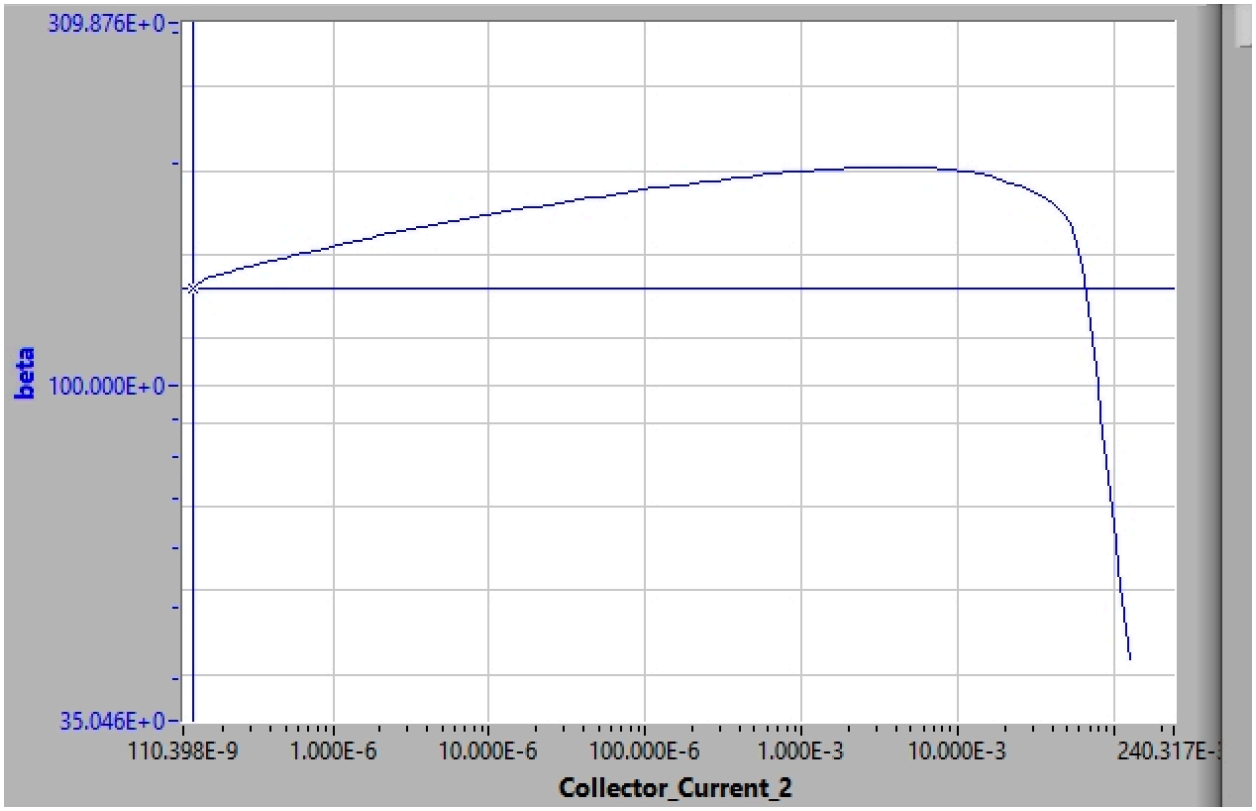
Here is the semilog Gummel plot:



Explanation:

This Gummel plot shows the exponential increase of the collector current and base current as functions of the base-emitter voltage in the BJT. The collector current initially rises faster, following an exponential relationship due to minority carrier injection, while the base current increases more gradually, due to carrier recombination. The ratio between the two currents reflects the transistor's current gain. Both currents are small at low base-emitter voltage but rise exponentially as voltage increases. Eventually the two will cross due to the decreased prominence of recombination at higher base voltages. High-level injection, where a large number of carriers are injected into the base, reduces the current gain and contributes to the bent shape of the curve.

Beta vs log(IC) log-log Plot



Explanation: This plot shows how beta (the current gain) increases steadily with increasing collector current until dropping dramatically at around 1e-2 amps of current. The initial increase in beta is due to how at low currents base recombination dominates, limiting gain. However, as current increases, the effects of recombination diminish and gain increases. This peaks when the BJT reaches its forward active region. However, then resultant current tapers off as seen in the Q2 graph. This causes the gain to decrease as a result due to high-level injection effects, where **higher carrier injection into the base and parasitic resistance reduce gain**. Increases in collector current require a much higher increase in base current relative to the previous parts of the curve. Overall, this explains the increasing behavior then stark drop in the above curve.

Q4. Find the subthreshold slopes of the collector and base currents, and their ideality factors. Do you see the Kirk effect at high current injection?

From the semilog graphs in the subsequent page, we can estimate a subthreshold slope.

For the collector current, we pick out a subthreshold slope of
0.551-0.484/decade=0.067V/decade

For the base current, we pick out a subthreshold slope of
0.5885-0.510/decade=0.0785V/decade

We can relate these slopes to the ideality factor. For subthreshold slope S, we know S * ln(10) = n*V_t, so the ideality factor can be computed accordingly. Doing this computation with the above subthreshold slopes slopes S. The collector current curve yields n = 1.1191434726, while the base current curve yields n = 1.31123526267

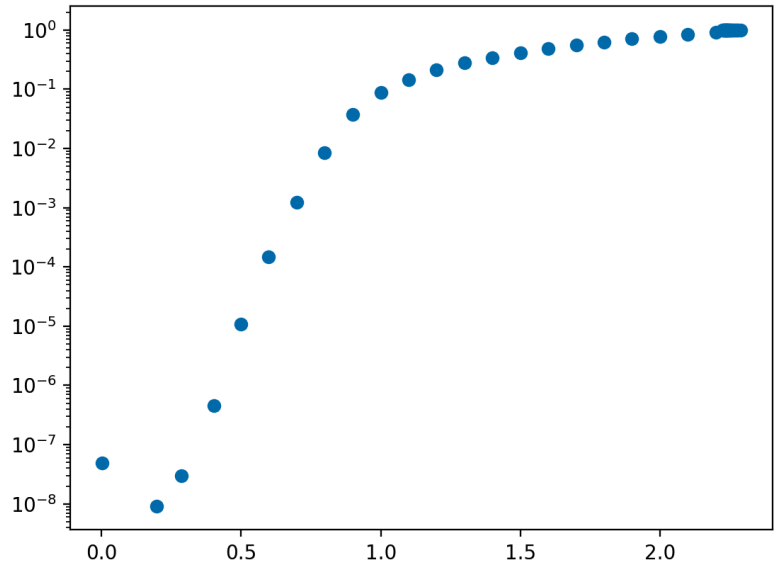
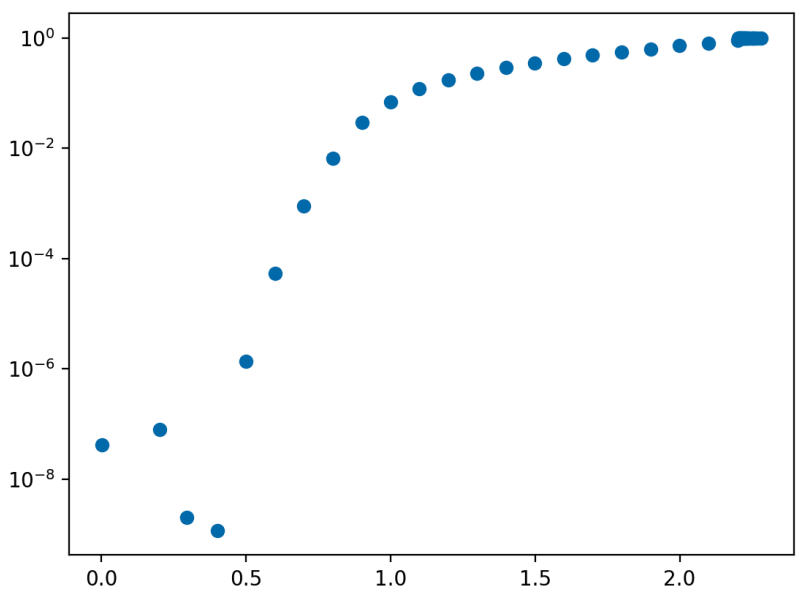
Collector Current Subthreshold Slope: 67mV/decade
Ideality Factor: n = 1.1191434726

Base Current Subthreshold Slope: 78.5mV/decade
Ideality Factor: n = 1.31123526267

Yes; I observe the Kirk effect at high current injection in the collector current plot. For our diode plots, at high voltage, you can see the curve start to plateau and data points start to bunch together at the top right end.

This is due to the Kirk effect (high injection effects) at very high currents.

Collector Current Plot (Amps vs Volts)



Base Current Plot (Amps vs Volts)